

PASSWORD STRENGTH CHECKER

Project Report

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1. Introduction

Passwords are the first line of defense in protecting digital accounts and sensitive information. Weak passwords can be easily guessed or cracked by attackers, leading to unauthorized access and security breaches. Therefore, it is important to ensure that users create strong and secure passwords.

This project, “Password Strength Checker”, is designed to evaluate the strength of a password based on predefined security criteria and classify it as Weak, Medium, or Strong.

2. Project Objective

The objective of this project is to develop a Python-based application that analyzes a user's password and determines its security level.

The program evaluates the password using the following criteria:

- Password length
- Presence of uppercase letters
- Presence of numeric digits
- Presence of special characters

Based on these checks, the program displays the overall password strength.

3. Problem Statement

Many users create passwords that are easy to remember but difficult to secure. Common passwords such as “123456” or “password” can be compromised within seconds.

This project aims to help users understand password security by providing instant feedback regarding the strength of their passwords.

4. Tools and Technologies Used

- Programming Language: Python 3
 - Development Environment: Visual Studio Code / Any Python IDE
 - Python Libraries:
 - string
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5. Project Methodology

The Password Strength Checker follows the following process:

1. Accept password input from the user.
 2. Check whether the password length is at least 8 characters.
 3. Verify the presence of uppercase letters.
 4. Verify the presence of numeric digits.
 5. Verify the presence of special characters.
 6. Calculate a score based on satisfied conditions.
 7. Classify the password as Weak, Medium, or Strong.
 8. Display the result to the user.
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6. Algorithm

Step 1: Start

Step 2: Take password input from the user.

Step 3: Check password length.

Step 4: Check for uppercase letters.

Step 5: Check for digits.

Step 6: Check for special characters.

Step 7: Calculate total score.

Step 8:

- If score $\leq 1 \rightarrow$ Weak Password
- If score = 2 or 3 $\rightarrow$ Medium Password
- If score = 4 $\rightarrow$ Strong Password

Step 9: Display password strength.

Step 10: End

7. Program Logic

The program evaluates four security conditions:

Security Check	Description
Length Check	Password should contain at least 8 characters
Uppercase Check	Password should contain at least one uppercase letter
Digit Check	Password should contain at least one number
Symbol Check	Password should contain at least one special character

Each successful condition contributes one point to the overall score.

8. Test Cases

Password	Result
abc	Weak
password	Weak
Daniyal123	Medium
Linux2026	Medium
Daniyal@123	Strong
Cyber#2026	Strong

9. Results

The developed application successfully evaluates password strength based on security requirements and provides immediate feedback to users.

The program correctly classifies passwords into Weak, Medium, and Strong categories according to the implemented validation logic.

10. Learning Outcomes

Through this project, the following concepts were learned:

- **Password security principles**
 - **Input validation**
 - **String handling in Python**
 - **Conditional statements**
 - **Basic cybersecurity practices**
 - **Security-focused programming**
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11. Future Improvements

The project can be enhanced by implementing:

- **Password entropy calculation**
 - **Detection of commonly used passwords**
 - **Password breach database integration**
 - **Graphical User Interface (GUI)**
 - **Real-time password strength meter**
 - **Password generation recommendations**
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12. Conclusion

The Password Strength Checker project successfully demonstrates the importance of password security and validation techniques. By evaluating passwords based on length, uppercase letters, digits, and special characters, the system helps users create stronger passwords and understand

fundamental cybersecurity concepts. This project provided practical experience in Python programming and introduced essential security validation techniques used in real-world applications.

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